25900649 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCSTA201 - ANALYTICAL TOOLS FOR MULTIVARIATE ANALYSIS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type questions

Answer any **7** questions

Each question carries **2** marks

1. State basis theorem of a vectorspace. [K] / [CO1]

2. State the condition for two vectors to be orthogonal. [K] / [CO1]

3. Explain the condition for a matrix A to be idempotent. [U] / [CO2]

4. State one difference between row echelon form and reduced row echelon form. [U] / [CO2]

5. State the condition for a matrix to be orthogonal matrix. [K] / [CO2]

6. Define the trace of a square matrix. [K] / [CO2]

7. What is the triangle inequality for norms? [K] / [CO1]

8. Write two applications of linear algebra in statistics? [U] / [CO1]

9. Define a diagonalizable matrix. [K] / [CO4]

10. Are eigenvectors of distinct eigenvalues linearly independent? [U] / [CO4]

[2x7 = 14]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **6** marks

11. a) Differentiate between scalar and vector with suitable examples. [A] / [CO1]
    b) Explain the significance of the norm of a vector.
    c) State the formula for calculating angle between the two vectors,u=(2,2,1) and v=(1,−1,2).

12. If a set of subspaces are given then show that the sum of subspaces is also a subspace of the vectorspace. [U] / [CO1]

13. $Verify\ that \det(AB) = \det(A)\det(B)\ , for\ A = \begin{bmatrix} 3 & 1 & 2 \\ 0 & -1 & 4 \\ 2 & 3 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 1 & 1 \\ 0 & 2 & 1 \end{bmatrix}.$ [A] / [CO2]

14. Determine the nature of the Quadratic form $Q = -2x^2 - 2y^2 - 5z^2 + 2xy + 4xz + 4yz.$ [U] / [CO2]

15. $a) State\ Cramers\ rule.$ [A] / [CO2]
    $b) Solve\ using\ Cramer's\ Rule:$
    $x + y + z = 2,\ 2x - y + z = 1,\ x - y + 2z = 4$

16. $Verify\ the\ Cayley - Hamilton\ theorem\ for\ A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ [A] / [CO4]

[6x4 = 24]

Part C

Essay type questions

Answer any **1** questions

Each question carries **12** marks

17. Define linear independence of vectors and examine the linear independence of the following vectors.. $\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \quad \vec{v}_3 = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}.$ [A] / [CO1]

18. $Let\ R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}\ and\ S = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}.$ [A] / [CO2]
    $Compute\ R(\frac{\pi}{4})SR(\frac{\pi}{2})$

[12x1 = 12]